

Ravulapalli Manas Venkata Sai

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Ashoka University, Sonipat, Haryana, India

RESEARCH INTERESTS

I work on normative and computational theory of how neural systems build structured internal models of the world, and on the same question in artificial systems. My thesis derives a *quotient law* for predictive learning: a recurrent predictive network trained on egocentric observations represents state space only up to the symmetry subgroup whose actions are indistinguishable from self-motion, which predicts when place fields should repeat across identical rooms. I am also interested in successor-style and factorised representations (TEM/SR lineage), and in the developmental question of when representational structure becomes usable during learning, including critical-period effects in language models and the gap between what a model represents and what it uses at query time. I am looking to do theory-first computational neuroscience in close contact with experimental data, not wet-lab work.

EDUCATION

Ashoka University

Sonipat, HR, India

B.Sc. (Hons.) Computer Science; minors in Biology and Psychology. GPA 3.66/4.00 Aug 2023 – May 2027 (expected)

- **Thesis (ISM):** *Cognitive maps are symmetry quotients: why place fields repeat in identical rooms.* Supervisor: Prof. Debayan Gupta.

Delhi Public School, Hyderabad, TG (Mathematics, Physics, Chemistry, Computer Science)

2022

PAPERS AND MANUSCRIPTS

- **Ravulapalli, M.** (2026). *Cognitive maps are symmetry quotients: why place fields repeat in identical rooms.* Single-author. Extended abstract submitted to **NeurReps 2026** (NeurIPS workshop); full manuscript in preparation for bioRxiv.
A predictive RNN trained on egocentric observations represents state space only up to the symmetry subgroup whose actions are indistinguishable from self-motion. Map quality degrades monotonically with the order of the landmark symmetry group (C_1, C_2, C_4); unit-level aliasing dissociates from population-level structure; head-direction input is required for global map stability; offline replay co-emerges with the fold at the same prediction horizon. The workshop version is framed around sensorimotor identifiability.
- **Ravulapalli, M., et al.** *The lifecycle of a capability: acquisition, retention, loss, and repair in continued pretraining.* In preparation, targeting ICLR 2027.
Preregistered campaign on OLMo-2 with frozen, hash-pinned predictions filed before launch. With zero occurrences of a capability's data in a 200M-token continued-training window, Adam's β_2 decides survival (0.95 kills, 0.999 preserves, 3/3 each); the ability to *acquire* the capability is installed by the stage-2 anneal rather than by stage-1 scale; the resulting failures are invisible to standard benchmarks (four non-target benchmarks move under 1.1 points) yet repairable in 64 updates.
- **Ravulapalli, M., et al.** *Visible but out-voted: in-context binding failures are query-time routing errors.* In preparation.
When a language model emits the wrong in-context binding, the correct binding remains decodable and causally effective; the failure localises to query-time attention routing and later-layer overwrite. A developmental analysis over 70 pretraining checkpoints (Pythia, OLMo-2) shows representation and use develop on different clocks.
- **Ravulapalli, M., et al.** *Load-dependent loss of plasticity in small language models.* In preparation.
~890 training runs on a controlled binding task. A formed-then-collapsed circuit is a worse (absorbing) state than one never formed (McNemar $p = 0.007$); a pulse-response plasticity measure decays 30 to 100× within ~8k steps of supervision deprivation; the critical window $\tau_c(K)$ shortens with binding load.

RESEARCH EXPERIENCE

Perseus (Y Combinator-backed)

Chief Scientist

Remote

2025 – Present

- Set and lead the company's research agenda across four lines: world models for code, developmental and continued pretraining, mechanistic interpretability, and evaluation. Supervise a small research team and external collaborations.

- **Coding world models.** Leading work on learned models of a codebase and its execution behaviour that an agent can roll forward and plan against, rather than retrieving text. Related architectural work (MapCore) adapts TEM-style structural priors as a persistent, slot-structured state on a transformer backbone.
- **Developmental pretraining.** Built the experimental program on when capabilities become acquirable, retainable, and recoverable during training, from a small controlled model up to OLMo-2 continued pretraining, under a preregistration protocol with predictions frozen before launch.
- **Interpretability.** Built the instrument stack for the routing and binding program (Jacobian-based binding probes, path patching, per-checkpoint developmental sweeps), and ran an adversarial audit of our own corpus that retracted several previously published numbers.
- **Evaluation.** Direct BioBench, an external-facing benchmark for biological mechanism reasoning built on a symbolic verifier that certifies or rejects a chain of reasoning rather than scoring accuracy, developed with external research partners including ARC.

Independent Research (ISM Thesis), Ashoka University

Sonipat, HR

Predictive RNNs and cognitive map formation

Aug 2025 – Present

- Implemented the Levenstein et al. (2024) sequential predictive RNN (NormReLU, rollout $k = 5$, RMSProp with per-parameter LR scaling) to test how environment statistics drive map formation.
- Built diagnostics for representational structure: occupancy-normalised mutual information, geodesic sRSA via Dijkstra for non-convex arenas, participation ratio, RDM Frobenius delta, and persistent homology via Ripser.
- Tested a topology-before-geometry hypothesis (whether correct Betti numbers emerge before metric structure) across ten environments varying geometry, topology, scale, and aliasing, in RatInABox and MiniGrid.

European Molecular Biology Laboratory (EMBL)

Heidelberg, Germany

Summer Student

Jul 2025

- Used optical tweezers to study how transcription factors locate specific DNA sequences, analysing kymographs to interpret binding dynamics; also worked with light-sheet and confocal platforms.
- Applied bio-image analysis, 3D reconstruction, and geometrical modelling to microscopy volumes, and evaluated how algorithm choice propagates into biological interpretation.

Neural Systems Lab, University of Hyderabad

Hyderabad, TG

Summer Research Assistant

Jun – Aug 2025

- Ran insect electrophysiology, including setup and calibration of the recording rig, and bulk drug tracing in grasshoppers for pathway visualisation.
- Designed and ran behavioural assays in weaver ants, including mirror-recognition replications; introduced BORIS and Ilastik to the lab's analysis workflow.

Neuro-ethology Lab, Ashoka University

Sonipat, HR

Research Assistant

Aug 2024 – Present

- Built an automated zebrafish tracking and movement-analysis tool (OpenCV, YOLOv8) to replace manual scoring, and closed-loop infrastructure for zebrafish VR.
- Developing a behavioural paradigm for numerical abilities in a zebrafish subpopulation; wrote the lab's recording and analysis protocols.

TRAINING

CAMP 2026, Computational Approaches to Memory and Plasticity

IISER Pune, India

Selected participant

2026

- Project on the goldfish oculomotor integrator (Seung et al.): NEURON simulation and dynamical systems analysis of persistent activity. Also implemented a Brian2 spiking Q-learning bandit and a Tsodyks–Markram dynamic synapse network.

TEACHING

Ashoka University

Sonipat, HR

Teaching Fellow, Computational and Theoretical Neuroscience

Jan – May 2025

- Co-designed the course with Prof. Bittu K. Rajaraman, delivered 15 lectures, and ran the journal club. Wrote MATLAB assignments on neural encoding and decoding, point and compartmental models, rate models, perceptrons, recurrent networks, and Hopfield networks.

Ashoka University

Sonipat, HR

Teaching Assistant, Introductory Electrophysiology

Oct – Nov 2025

- Designed a hands-on course for 10 students on setup, maintenance, and error checking of an Axon rig for patch clamp and microelectrode recording, with ERG/EAG practicals and Python spike train analysis tutorials.

SELECTED PROJECTS

- **Grid field distortion under reward modulation:** Continuous attractor models of how reward gradients perturb grid fields. Measured grid score, spacing, and orientation shifts, and used persistent homology to characterise disruption of the toroidal population manifold.
- **Zebrafish collective behaviour:** Tracking pipeline for shoaling distance, velocity, thigmotaxis, and turning-angle distributions, with agent-based simulations quantifying polarisation and nearest-neighbour distributions.
- **Cognitive assessment platform for neurodegenerative disorders:** Digital battery for attention, working memory, and motor deficits in Alzheimer's patients and controls, using convex hull estimation and constrained path tracing to probe visuospatial processing, with automated scoring and longitudinal logging.

TECHNICAL SKILLS

Modelling and theory: Recurrent and predictive networks, continuous attractor networks, reinforcement learning, representational geometry (RSA, participation ratio, RDMs), persistent homology, dynamical systems analysis, transformer interpretability (probing, path patching)

Programming: Python, C++, C, Rust, MATLAB, JavaScript, C#, Assembly (MIPS, RISC)

Scientific computing: PyTorch, NumPy, SciPy, statsmodels, PyMC, Brian2, NEURON, NEST, Elephant, RatInABox, MiniGrid, Ripser, Git, Docker

Statistics: Regression, mixed-effects models, Bayesian inference, survival analysis, pre-registered hypothesis testing, bootstrap and cluster-aware inference, power analysis

Neural data analysis: SpikeInterface, Kilosort, Open Ephys, Suite2p, SIMA, DeepLabCut, SLEAP, Napari; SPM for voxelwise fMRI statistics with fMRIPrep and nilearn for ROI and connectivity modelling

Experimental: Patch clamp, intra- and extracellular microelectrode recording, LFP, ERG/EAG, rig setup and maintenance, dissection, bulk drug tracing, behavioural paradigm design

Microscopy and imaging: Light sheet, confocal, fluorescence, optical and scanning electron microscopy, optical tweezers; ImageJ/Fiji, Ilastik, BORIS, kymograph interpretation, 3D reconstruction

Systems: Closed-loop experimental systems, ML inference pipelines, RESTful APIs, Node.js, React, Tauri, SQL (MySQL, PostgreSQL), MongoDB, AWS

REFERENCES

Available on request.